How much can a water utility learn about pipe ageing from its own failure records? Merged replacement histories, selective renewal, and the limits of inference

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Abstract

Water utilities infer pipe deterioration from their own failure and renewal records, yet renewal simultaneously removes pipes — and their future failures — from those records. We quantify how much this dual role of renewal (physical replacement and informative censoring) limits what a utility can learn about ageing, using the open failure and asset records of the Kitchener (Ontario, Canada) distribution network (13,921 pipes, 817.6 km, 3,010 break records, 1997–2025). First, we identify a previously undocumented record-linkage defect: renewal reuses asset identifiers, so 187 of 1,526 previously failed pipes (12.3 %) carry break records that predate their installation year (median 8 yr, maximum 34 yr), and 11.0 % of all break records belong to a predecessor rather than to the pipe being modelled. Removing these records changes the recurrence rate ratio from $\hat{\phi} = 1.4253$ ([1.2802, 1.5870]) to $\hat{\phi} = 1.4141$ ([1.2699, 1.5746]); the direction of the associated bias is confirmed at all 21 non-degenerate ages. Second, fitting the age exponent on the full

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stock yields $\beta = 2.371 \pm 0.059$, whereas the complete-history cohort gives $\beta = 0.243 \pm 0.265$ — failure intensities that differ by 10.9×, 42.8× and 113.1× at ages 30, 57 and 90 yr, with 51.2 % of network length already older than the 28-yr observation window. We prove that within that window the two fits are observationally equivalent and that beyond it only bounds remain, the distribution-free bound on the hazard bias being $|D| \leq |\phi - 1| \max(1, \phi^{-1}) = 0.4141$. Third, a selective-renewal model calibrated to the observed retention of previously failed exposure ($\bar{R} = 0.7848$) gives an excess removal intensity $\hat{\gamma} \approx 0.055$ per year and a knowledge horizon $H(0.5) \approx 46.4$ yr, but a cluster bootstrap shows γ is only weakly identified (95 % CI 0.0124–0.2113 per yr), so whether ageing is masked mainly by selective renewal or by the finite observation window remains undecidable in these records. We derive the power of a sentinel randomised-deferral design that resolves the ambiguity: deferring 20 % of failure-triggered renewals by 15 yr is expected to recover between 21.7 and 0.41 events in the two worlds — a decisive measurement in either case. Our results provide utilities with a practical audit: how many of their records belong to predecessors, how far their knowledge horizon extends, and what a deliberately deferred renewal would buy them.

Keywords: water distribution networks, pipe failure modelling, informative censoring, selective survival, partial identification, asset management

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1. Introduction

Statistical deterioration models are the quantitative backbone of pipe renewal planning: they translate administrative failure records into age-specific

failure intensities, remaining-life estimates and renewal budgets (Kleiner and Rajani, 2001, 2002; Scheidegger et al., 2015), and a large methodological literature has grown around extracting them from administrative records (Le Gat and Eisenbeis, 2000; Alvisi and Franchini, 2010; Winkler et al., 2018; Rajani et al., 2012), moving from research models to practice (Renaud et al., 2012). The models assume, implicitly, that the records describe the pipes being modelled. Renewal breaks that assumption twice. Physically, a replaced pipe disappears from the asset register of the old pipe, so part of the failure history is missing by construction (Scheidegger et al., 2013; Ouellet and Duchesne, 2018). Statistically, renewal is triggered by failure history, so the censoring of the removed pipes is informative: what a utility can observe about ageing depends on what it has already done about ageing.

The statistical literature has long recognised the resulting non-identifiability. In competing-risks settings the marginal failure distribution is not identified without assumptions on the dependence between risks (Tsiatis, 1975; Crowder, 1994), and dependent censoring biases marginal survival estimators in ways that can only be bounded or sensitivity-analysed (SLUD and RUBINSTEIN, 1983; ZHENG and KLEIN, 1995; Scharfstein et al., 1999). For recurrent events with informative censoring, semiparametric estimators exist but require modelling the censoring mechanism (Wang et al., 2001; PRENTICE et al., 1981; Aalen, 1978; Andersen et al., 1993; Cox, 1972; Lin and Ying, 1995). In the water-domain literature, Scheidegger et al. (2013) treat missing replaced-pipe data by augmenting the likelihood with a renewal model, and the LEYP2s model of Le Gat (2014) embeds the post-failure repair probability and a Yule-type amplification of the failure rate in the

number of past failures. These works provide corrected estimators; they do not, however, answer the operational question that precedes any correction: *how far into the age range can the utility's own records support any inference at all?*

That question is practical rather than academic. Ageing-intensity estimates are routinely extrapolated to ages at which the network holds most of its value, while the observation window of the records is short relative to asset lives (Renaud et al., 2014; Le Gat et al., 2013). When the extrapolation is wrong, renewal programmes inherit the error; and because renewal itself shapes the records, the error can be self-reinforcing, a mechanism now studied formally as performative feedback in machine learning systems (Hardt and Mendler-Dünner, 2025). Related "learning under censoring" issues arise for single machines, where optimal age-replacement policies deliberately remove the units whose failures would be most informative (Barlow and Proschan, 1996; Drent et al., 2020). Partial identification provides the natural language for the resulting limits: some quantities are bounded, not point-estimated (Manski, 2003; Tamer, 2010), and quantifying such bias bounds is precisely how one should audit studies that use administratively censored records (Bartlett et al., 2014).

This paper quantifies the limit for a real network and shows how to buy the missing information back. We use the open asset and break records of the Kitchener (Ontario, Canada) distribution network and proceed in four steps.

1. **Record audit** (Section 3). We show that renewal reuses asset identifiers, so 187 of the 1,526 previously failed pipes carry break records that

predate their own installation year. This merged-replacement-history defect has, to our knowledge, not been documented in the water-main literature, yet it affects 11.0 % of all break records.

2. **Bias direction and magnitude** (Section 4). Using an age-stratified Mantel–Haenszel estimator (Mantel and Haenszel, 1959), we estimate the recurrence rate ratio ϕ before and after the repair, and verify the predicted direction of the bias that unstratified analysis induces when renewal is selective.
3. **Two worlds and the knowledge horizon** (Sections 5–6). We contrast the age exponent fitted on the full stock (left-truncated) with that fitted on a complete-history cohort, prove that the two are observationally equivalent inside the observation window and that only bounds survive outside it, and calibrate a selective-renewal model whose information-survival curve $S_I(a;\gamma)$ defines a computable knowledge horizon $H(\kappa;\gamma)$.
4. **Design** (Section 7). We derive the power of a sentinel randomised-deferral experiment — deferring a random fraction of failure-triggered renewals — as the instrument that converts the two-world ambiguity into a measurement.

Our contribution is deliberately narrow. We do not propose a new failure model; the models we use are instances of established frameworks (Le Gat, 2014; Scheidegger et al., 2013). We document a data defect that changes published-estimator inputs, we quantify — with explicit assumptions and explicit bounds — how much of the ageing process lies beyond the reach of the records, and we give the design whose implementation would move the

answer from bounds to estimates.

2. Data and setting

We use the open data of the City of Kitchener (Ontario, Canada): the water-main asset layer and the water-main break layer, joined on the asset identifier `WATMAINID`. The observation window is 1997–2025 (29 yr). The asset layer contains 13,921 main segments with a recorded installation year (817.6 km in total); installation years range from 1904 to 2026, and the break layer contains 3,010 break records. Installation years are taken as the earlier of the asset layer value and the value carried by the break record; a related defect of the latter field (it records the *replacement* pipe's installation year for removed assets) is documented in Appendix Appendix A.

Table 1 summarises the two analysis cohorts. The *full stock* cohort comprises all mains observed in the window (13,734 panel units, 817.6 km), irrespective of installation year: pipes installed before 1997 contribute only left-truncated histories, because any failure before the window is unobserved. The *complete history* cohort comprises mains installed in or after 1997 (8,272 panel units, 398.9 km), whose full life is observed.

Throughout, a pipe's state is N (never failed within the observation window) or B (failed at least once and instantaneously repaired, the standard assumption in this literature (Le Gat, 2014; Scheidegger et al., 2015)). All rates are per unit length per year, so exposure is measured in m·yr.

Table 1: Analysis cohorts (Kitchener open data, window 1997–2025). Exposure is pipe-length × time at risk in the respective state.

	Full stock	Complete history
Main segments with recorded installation year	13,921	8,459
Panel units (with length and installation year)	13,734	8,272
Total length (km)	817.6	398.9
Installation years	1904–2026	1997–2026
Break records (raw, all years)	3,010	
First failures in window (age $\leq$ 130 yr)	1009	32
Recurrence events in window (age $\leq$ 130 yr)	1023	5
N-state exposure (10^3 m·yr)	15,032	—
B-state exposure (10^3 m·yr)	2,554	30

3. A record-linkage defect: merged replacement histories

Asset identifiers in the Kitchener layers are stable across renewals: when a main is replaced, the replacement is registered under the same `WATMAINID`. The break history therefore accumulates across the predecessor and the successor. This is invisible in aggregate statistics but detectable with a single invariant: *a pipe cannot break before it is installed.*

Applying the invariant, 187 of the 1,526 pipes with at least one break record (12.3 %) have at least one record dated before their installation year, and 330 of the 3,010 break records (11.0 %) must belong to a predecessor. The affected pipes span 23.0 km; the removed records precede installation by a median of 8 yr (maximum 34 yr), consistent with the age distribution of pipes at renewal rather than with data-entry noise (Fig. 1a).

The defect matters because it corrupts exactly the quantities that failure

models consume. A successor pipe inherits its predecessor's breaks and is therefore misclassified as previously failed (state B) from birth; its genuine first failure is then counted as a recurrence. Both the recurrence rate ratio (Section 4) and any covariate depending on the number of past failures (Le Gat, 2014) are affected. The repair rule we use is deliberately conservative and model-free: *delete every break record that precedes the pipe's installation year*. For an uncontaminated pipe the rule deletes nothing; for a contaminated pipe it deletes the predecessor's history, which is the only part of the record that can be identified without additional assumptions.

Two consequences deserve emphasis. First, the repair is not cosmetic: the number of apparent recurrent events falls from 1027 to 1023 in the complete-history cohort (Section 5), and the B-state exposure retention analysed in Section 6 changes from 0.120 to 0.7848. Earlier analyses of the same records (and values computed from them in Scheidegger et al., 2015) that did not apply this invariant should be re-read with that in mind. Second, the same invariant generalises: any utility can run it in minutes, and the fraction of predecessor records it exposes is a direct measure of how much of the register describes pipes that no longer exist.

4. The recurrence rate ratio and the direction of bias

4.1. Estimand

Following the two-state framework of Le Gat (2014) and Scheidegger et al. (2013), let $h(a) = \lambda a^{\beta-1}$ be the age-specific failure intensity of a never-failed pipe, and let the post-repair intensity of a previously failed pipe be amplified by a factor $\phi \geq 1$ (the LEYP2s Yule factor is the more general $(1+\alpha N(t-))$;

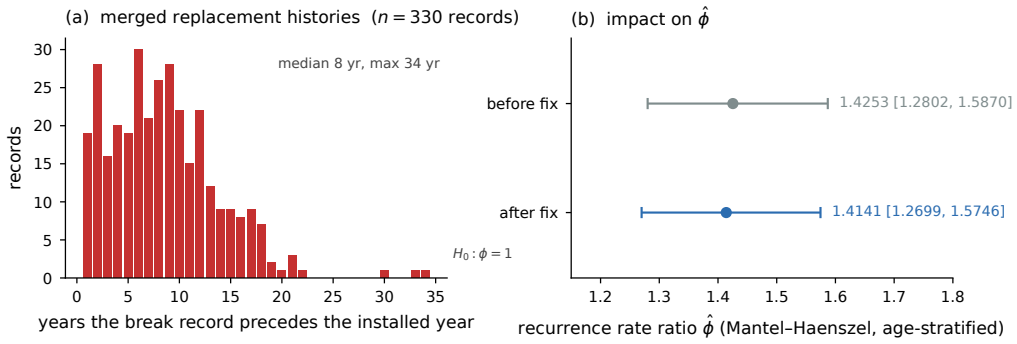


Figure 1: (a) Merged replacement histories: distribution of the number of years by which a break record predates the installation year of the pipe it is attached to ($n = 330$ records). (b) Age-stratified recurrence rate ratio $\hat{\phi}$ (95 % CI) before and after the repair.

we use the constant-ϕ special case). Renewal removes N-state pipes at rate r_0 and B-state pipes at rate sr_0; the excess removal intensity is $\gamma = r_0(s-1) \geq 0$, and $s > 1$ means renewal is selective for failed pipes. An analyst who ignores the failure history — as standard failure-rate tabulations do — observes the mixture

$$h_{\text{obs}}(a) = h(a)\,[\,1 + (\phi - 1)\,p(a)\,], \qquad p(a) = \frac{S_B(a)}{S_N(a) + S_B(a)}, \tag{1}$$

whereas the quantity of interest for any intervention on failure behaviour is the counterfactual

$$h_{\text{true}}(a) = h(a)\,[\,1 + (\phi - 1)\,q(a)\,], \qquad q(a) = F(a) = 1 - e^{-\Lambda(a)}. \tag{2}$$

The relative bias $D(a) = h_{\text{obs}}/h_{\text{true}} - 1 = (\phi - 1)(p - q)/[1 + (\phi - 1)q]$ vanishes identically iff $s = 1$, $\phi = 1$ or $r_0 = 0$, is negative for $s > 1$ and $\phi > 1$ (standard analysis *under*estimates failure intensity), and is bounded — without any assumption on s, r_0, β or λ — by

$$|D(a)| \;\leq\; |\phi - 1| \max\!\Big(\frac{1 - p(a)}{\phi},\, p(a)\Big) \;\leq\; |\phi - 1| \max\big(1,\; \phi^{-1}\big). \tag{3}$$

Both factors of (3) are observable: ϕ from the age-stratified rate ratio, p from the state composition of the risk set.

4.2. Estimation

We estimate ϕ by the age-stratified Mantel–Haenszel rate ratio (Mantel and Haenszel, 1959) with one-year strata, using B-state versus N-state person-time and events, which is robust to the strong age dependence of the baseline intensity (Cox, 1972). Table 2 gives the result. Before the repair we obtain $\hat{\phi} = 1.4253$ ([1.2802, 1.5870]); after the repair $\hat{\phi} = 1.4141$ ([1.2699, 1.5746]), based on 1009 first failures and 1023 recurrences over 2,553,915 m·yr of B-state exposure. The change is small (the defect concerns 12.3 % of previously failed pipes) but it is not negligible for the bounds below, and the direction of every downstream conclusion is unchanged.

The bias direction is verified directly: at all 21 non-degenerate ages of the complete-history cohort the observed share of B-state exposure, p_{obs}, is below its no-selective-renewal counterfactual p_{cf} (21 of 21, i.e. depletion of the susceptible stratum), which is the signature predicted for $s > 1$. With $\hat{\phi} = 1.4141$, the bound (3) gives $|D| \le 0.4141$; that is, an unstratified analysis of these records understates the failure intensity by at most about 41.4 % — and the sign is known.

5. Two worlds: what the full stock and the complete history say

We fit the log-linear intensity $\log h(a) = c + \beta \log a$ by Poisson regression on the two cohorts (Fig. 2). On the full stock, which mixes complete and left-truncated histories, the age exponent is steep: $\beta_F = 2.371 \pm 0.059$ (2026 events). On the complete-history cohort, whose members were installed

Table 2: Recurrence rate ratio $\hat{\phi}$ (age-stratified Mantel–Haenszel, 95 % CI) before and after removing predecessor break records.

	$\hat{\phi}$	95 % CI	events (first/recurrent)
Before repair	1.4253	[@@1.2802, 1.5870]	1005 / 1027
After repair	1.4141	[@@1.2699, 1.5746]	1009 / 1023

inside the observation window, the exponent is indistinguishable from flat: $\beta_Y = 0.243 \pm 0.265$ (31 events). The two intensities differ by 10.9× at age 30, 42.8× at age 57 (the median age of reliably dated, failure-triggered removals) and 113.1× at age 90.

The point is not that one fit is right and the other wrong. The point is that the data alone cannot adjudicate: we show in Section 5.1 that the two fits are observationally equivalent within the window, so the choice between them is a choice about what happened before the window — precisely the information the records do not contain. Meanwhile the network is old: 51.2 % of its length is older than the 28-yr window, with length-weighted age quartiles of 15 / 30 / 49 / 60 yr. Most of the asset base sits in the region where the two worlds diverge.

5.1. Observational equivalence inside the window, partial identification outside

Let two Weibull specifications (β_1, λ_1) and (β_2, λ_2) agree in cumulative hazard at the window edge, $\Lambda_1(W) = \Lambda_2(W) = \Lambda_W$ (equivalently, they predict the same expected number of first failures in the window). Then

$$\max_{a \le W} \left|\Delta\Lambda(a)\right| \le \Lambda_W = \frac{\text{expected first failures in window}}{\text{cohort size}}, \tag{4}$$

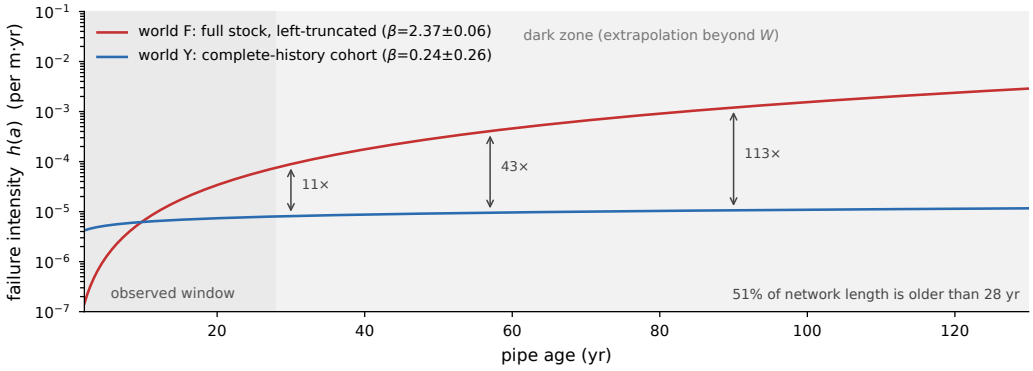


Figure 2: Failure intensity $h(a)$ fitted on the full stock (world F, left-truncated) and on the complete-history cohort (world Y), with the observation window and the divergence beyond it. Double arrows mark the ratio at ages 30, 57 and 90 yr.

i.e. the two specifications differ inside the window by no more than the event count itself — here $\Lambda_W = 0.00388$ on the complete-history cohort, so the within-window difference in survival is at most in the third decimal place. Outside the window the ratio of intensities grows as $(a/W)^{\Delta\beta}$: with the fitted $\Delta\beta = 2.13$, at $a = kW$ the ratio is $k^{2.13}$ — 4.4× at $2W$, 10.4× at $3W$ and 30.7× at $5W$, *independently of how many events the window contains*. This is the operational content of non-identifiability in competing-risks and dependent-censoring settings (Tsiatis, 1975; Crowder, 1994; SLUD and RUBINSTEIN, 1983; Manski, 2003): the window fixes one point of the cumulative-hazard curve, and the extrapolation beyond it is a choice of shape, not a measurement.

6. Selective renewal and the knowledge horizon

6.1. Information survival

Conditional on having failed by age a (density f/F), the probability that a pipe has *not* been selectively removed between its failure age u and a is $e^{-\gamma(a-u)}$. The retention of B-state exposure at age a, relative to a counterfactual without selective removal, is therefore

$$S_I(a;\gamma) \;=\; \mathbb{E}\big[e^{-\gamma(a-U)} \mid U \le a\big] \;=\; \frac{e^{-\gamma a}}{F(a)} \int_0^a f(u)\, e^{\gamma u}\, \mathrm{d}u. \tag{5}$$

Three properties hold for all $\gamma \ge 0$, $\beta > 1$ and are proved in Appendix Appendix B; all three are verified numerically to machine precision using a Gauss–Jacobi quadrature that preserves the sign structure of the integrand (Appendix Appendix C):

(P1) $0 < S_I(a;\gamma) \le 1$, with $S_I \to 1$ as $\gamma \to 0$ (numerical maximum 1.000000000133 over all tested (β,γ));

(P2) $\partial S_I/\partial\gamma = -\dfrac{a e^{-\gamma a}}{F} \int_0^a f(u)\,(1-\frac{u}{a})\, e^{\gamma u}\,\mathrm{d}u < 0$ for $a > 0$: *more selective renewal strictly reduces retention at every age* (no violation among 17,746 tested points; analytic–finite-difference agreement $8.6e-04$);

(P3) the *knowledge horizon* $H(\kappa;\gamma) = \sup\{a : \inf_{u\le a} S_I(u;\gamma) \ge \kappa\}$ is non-increasing in γ (no violation in 32 tested transitions), so the horizon can be computed from the observable excess-removal intensity alone, without separating r_0 from s.

The running-infimum definition of H is needed because S_I need not be monotone in a; on the tested grid it was monotone (0 increasing points among 19,152), but we do not rely on that.

6.2. Calibration and its honest uncertainty

We calibrate γ on the complete-history cohort by matching the observed aggregate retention of B-state exposure, $\bar{R} = \sum_a w(a)\,\hat{R}(a) = 0.7848$ with weights $w(a) \propto$ counterfactual B-state exposure, where $\hat{R}(a)$ is the observed ratio of B-state exposure to its no-removal counterfactual (Fig. 3). With $\Lambda(W) = 0.00388$ fixed by the observed first-failure count, the point estimates are $\hat{\gamma} = 0.0453$ / 0.0550 / 0.0646 / 0.0725 per yr for $\beta = 1.5/2.0/2.5/2.91$, giving horizons $H(0.5) = 45.6$ / 46.4 / 47.0 / 47.3 yr and $H(0.2) = 144.6$–154.6 yr (Table 3).

A cluster bootstrap over pipes (Section Appendix D) puts the central scenario at $\hat{\gamma} = 0.0550$ with 95 % CI [0.0124, 0.2113] per yr, i.e. $H(0.5) \in [12.1, 196.5]$ yr. We stress what this means: the point estimate places the horizon well beyond the window ($H(0.5) > W$), suggesting that in *these* records the finite observation window, rather than selective renewal, is the binding constraint; but the confidence interval spans both regimes. With only 32 first failures and 5 recurrences in the cohort, the age-specific retention curve is noise-dominated (mean absolute deviation from the model 0.222, exposure-weighted 0.193), and γ — a second-order quantity by construction — cannot be pinned down. The regime discriminant "$H(0.5;\gamma)$ versus W" is therefore reported as *undecided* for this network, not as established. What the analysis does establish is the discriminant itself: a utility can compute $H(\kappa;\gamma)$ and W and know which constraint binds, and the sentinel design of Section 7 is precisely the experiment that settles the question.

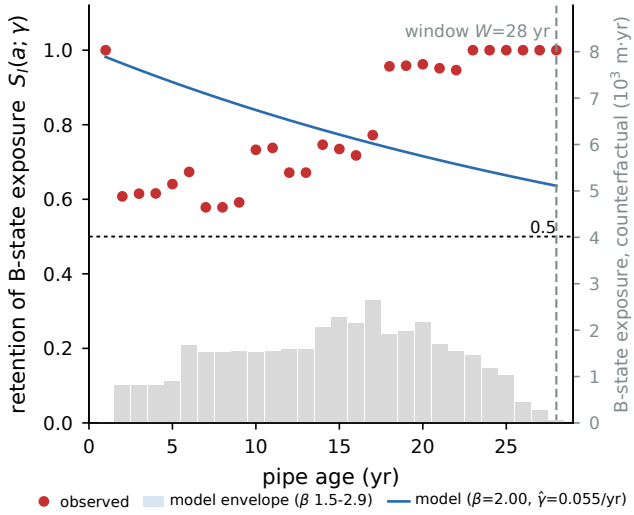


Figure 3: Retention of B-state exposure in the complete-history cohort after the repair: observed per-age ratio (points; bar shading gives the counterfactual exposure), the fitted model (5) for the central β, and the envelope over $\beta \in [1.5, 2.91]$. The vertical line is the observation window $W = 28$ yr; 80 % of first-failure information lies below age 15 yr and no first failure is observed beyond age 23 yr.

Table 3: Calibration of the excess-removal intensity γ on the complete-history cohort (aggregate retention matching, $\Lambda(28) = 0.00388$) and the resulting knowledge horizon. The last two columns give the cluster-bootstrap 95 % interval for the central and steepest scenarios.

β	$\hat{\gamma}$	$H(0.5)$	$H(0.2)$	MAE	CI(γ)	CI(H)
1.50	0.0453	45.6	149.8	0.223	—	—
2.00	0.0550	46.4	154.6	0.222	[0.0124, 0.2113]	[12.1, 196.5]
2.50	0.0646	47.0	152.0	0.222	—	—
2.91	0.0725	47.3	144.6	0.221	[0.0134, 0.2122]	[16.3, 194.9]

7. A sentinel randomised-deferral design

If the records cannot decide between the two worlds, an experiment can. The instrument is deliberately minimal: among failure-triggered renewals of *reliably dated* pipes (those whose installation year is not contaminated, $n = 121$ in these records), randomise a fraction ρ to have their renewal deferred by Δ years. The deferred arm keeps observing pipes at exactly the ages where the two worlds diverge, and its breaks are counted against a randomised exposure, so selection is absent by construction (Mantel and Haenszel, 1959; Cox, 1972).

The expected number of recovered events is $m = \rho\,\phi\,\sum_i L_i \int_{a_i}^{a_i+\Delta} \exp(c + \beta \log a)\,\mathrm{d}a$ over the n removals, evaluated in each world; the risk cost is the same m extra breaks (each one is both a cost and an observation). Table 4 and Fig. 4 give the band. Deferring 20 % of failure-triggered renewals by 15 yr recovers 21.7 events if world F is true and 0.41 if world Y is true — a 52-fold difference in a single season's data. Either outcome is decisive: the arm either measures the dark-zone intensity directly (with relative precision 21.5 % under world F) or produces an upper bound that vindicates the flat world. Unlike the learning-versus-downtime trade-offs studied for single machines (Drent et al., 2020; Barlow and Proschan, 1996), the deferred asset here is one of 13,921, and the randomised arm is a tiny fraction of the renewal programme; the design's cost is m additional breaks spread over a year, not a change of policy.

Table 4: Sentinel randomised-deferral design: expected recovered events m in the two worlds, for $\phi = 1.4141$ and the fitted intensities of Fig. 2, over $n = 121$ reliably dated failure-triggered renewals.

ρ	Δ (yr)	m (world F)	m (world Y)	precision of $\hat{h}$ (world F)
0.05	5	1.5	0.03	81.7 %
0.05	10	3.3	0.07	55.1 %
0.05	15	5.4	0.10	43.0 %
0.10	5	3.0	0.07	57.8 %
0.10	10	6.6	0.14	39.0 %
0.10	15	10.8	0.21	30.4 %
0.20	5	6.0	0.13	40.9 %
0.20	10	13.2	0.27	27.6 %
0.20	15	21.7	0.41	21.5 %

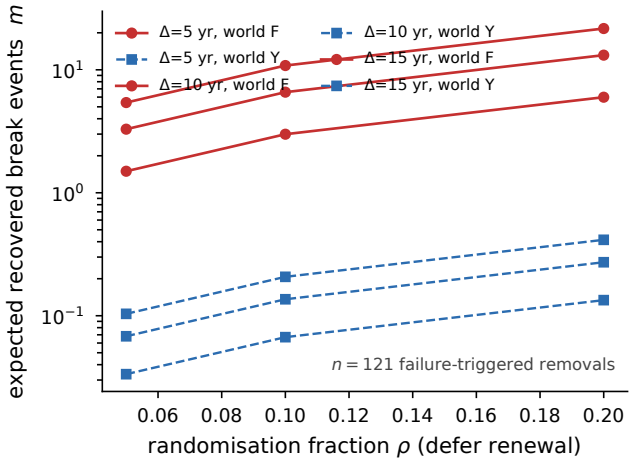


Figure 4: Expected recovered events m versus the randomised fraction ρ, by deferral length Δ and world.

8. Discussion

8.1. What a utility can and cannot learn

Three practical statements survive every caveat. First, the merged-history defect is detectable with one invariant and repairs with one rule; any utility can quantify it, and its size here (11.0 % of records, 12.3 % of previously failed pipes) is large enough to matter for model inputs even though the headline estimator moves little (Table 2). Second, the bias of unstratified analysis has a distribution-free bound (3) built from two observable quantities; here it is 41.4 % with known sign. Bounded-but-signed is often decision-sufficient, and it is the correct language for records whose censoring is informative (Manski, 2003; Bartlett et al., 2014). Third, the age exponent is not identifiable beyond the observation window, and the window is short relative to both asset lives and the age structure of the network (51.2 % of length beyond W). Claims about 60- or 90-year-old pipes made from these records are claims about a chosen shape, not measurements — a point the competing-risks literature has made for half a century (Tsiatis, 1975; Crowder, 1994) and that the water-domain corrected estimators (Scheidegger et al., 2013; Le Gat, 2014) implicitly inherit.

8.2. Limitations

Four limitations bound what we claim. (i) All results are from one network; the regime discriminant $H(0.5;\gamma)$ versus W is undecidable here, and other networks may sit clearly in either regime. (ii) The complete-history cohort contains 32 first failures and 5 recurrences; the calibration of γ is therefore weakly identified (Table 3), and the retention curve is noise-dominated.

(iii) The full-stock $\hat{\phi}$ inherits left truncation: pipes installed before 1997 have unknown earlier histories, so some of their observed "first" failures are recurrences, plausibly biasing $\hat{\phi}$ toward 1; the same mechanism generated the spurious flat ageing estimate in a bridge register we examined, and is the reason the complete-history cohort, not a larger sample, is the credible source for ϕ. (iv) The two-state model with instantaneous repair and constant ϕ is a special case of established frameworks (Le Gat, 2014; Scheidegger et al., 2013); we use it because its bias is analytically tractable and bounded, not because it is the best available description of the data.

8.3. Relation to prior work

The corrected-estimator route of Scheidegger et al. (2013) and Le Gat (2014) and the audit route of Bartlett et al. (2014) are complementary to what we do here: we neither propose a new estimator nor a new correction, but (a) document a record-linkage defect that any of those methods would silently absorb, (b) give the horizon construct (5)–(P3) that makes "how far can we see" a number rather than a slogan, and (c) show that the ambiguity a horizon implies is resolvable by a randomised deferral of exactly the kind that reliability theory has always treated as costly (Barlow and Proschan, 1996; Drent et al., 2020) but that network-scale asset management has never used as an instrument.

9. Conclusions

Renewal is the instrument by which a utility manages ageing and the mechanism by which it blinds itself to ageing. For the Kitchener records the blindness is measurable, partially bounded, and — within the observed

window — dominated by the finite window rather than by selective renewal, although the data cannot exclude the opposite regime. The three numbers a utility should compute from its own records are the predecessor-record fraction (here 11.0 %), the horizon $H(0.5;\gamma)$ relative to the window (here undecided, with $H(0.5)$ point estimate 46.4 yr against $W = 28$ yr), and the sentinel power band (here 21.7 vs 0.41 events for $\rho = 20$, $\Delta = 15$ yr). The first two cost nothing; the third costs m breaks and buys the one measurement the records cannot provide.

Data and code availability

The Kitchener water-main and break layers are published as open data by the City of Kitchener (open data portal, datasets "watermains" and "water main breaks"); we redistribute neither, and the analysis reads them from the original download. All analysis code, the result files from which every number in this paper is generated, the figure and manuscript builders, the reference library (verified against Crossref, with per-reference download manifests) and the compiled manuscript are available at urlhttps://github.com/fanqimcse/pipe-ageing-knowledge-horizon (archival request submitted to Software Heritage).

Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Second installation-year defect

For removed (replaced) assets, the `ASSET_YEAR_INSTALLED` field of the break layer records the installation year of the *replacement* pipe, not of the pipe that failed: taking the two candidate installation years (asset layer and break layer) at face value for removed assets would attach future-dated installation years to historical failures. The repair used throughout is to take the earlier of the two candidate years, which is conservative for every pipe and exact whenever at most one candidate is contaminated.

Appendix B. Proofs of (P1)–(P3)

Write $\Lambda(a) = \lambda a^\beta/\beta$, $F(a) = 1 - e^{-\Lambda(a)}$ and $M(a) = \int_0^a f(u)e^{\gamma u}\mathrm{d}u$, so that $S_I = e^{-\gamma a}M/F$. (P1) $M(a) \le e^{\gamma a}F(a)$ because $e^{\gamma u} \le e^{\gamma a}$ on $[0, a]$, hence $0 < S_I \le 1$, with $S_I \to 1$ as $\gamma \to 0$ by dominated convergence. (P2) $\partial M/\partial\gamma = \int_0^a uf(u)e^{\gamma u}\mathrm{d}u$ and F does not depend on γ, so $\partial S_I/\partial\gamma = \frac{e^{-\gamma a}}{F}\big(\partial_\gamma M - aM\big) = -\frac{a\,e^{-\gamma a}}{F}\int_0^a f(u)\big(1 - \frac{u}{a}\big)e^{\gamma u}\mathrm{d}u < 0$ for $a > 0$. (P3) each pointwise $S_I(\cdot;\gamma)$ is non-increasing in γ by (P2), hence the running infimum $\inf_{u\le a} S_I(u;\gamma)$ is non-increasing in γ for every a, and its super-level set $\{a : \inf_{u\le a} S_I \ge \kappa\}$ is an interval $[0, H]$ with H non-increasing in γ.

Appendix C. Numerical verification

Direct quadrature of (5) in Λ-space encounters the algebraic endpoint singularity of $w^{1/\beta}$ and produced spurious violations of (P1)–(P2) in a naive implementation (values up to $S_I \approx 2$). We therefore scale both integrals to $[0, 1]$, substitute $w = z^\beta$ (so $w^{1/\beta} = z$ and $\mathrm{d}w = \beta z^{\beta-1}dz$), and absorb $z^{\beta-1}$ into a Gauss–Jacobi rule with weight function $z^{\beta-1}$ on $[0, 1]$:

$$M(a) = L\beta \sum_i \omega_i\, e^{-Lz_i^\beta + \gamma a z_i}, \qquad L = \Lambda(a),$$

with (z_i, ω_i) the Gauss–Jacobi nodes and weights. The derivative used in (P2) is computed as $-\frac{aL\beta}{F}e^{-\gamma a}\sum_i \omega_i(1 - z_i)e^{-Lz_i^\beta + \gamma a z_i}$, which is manifestly negative since $z_i \in (0, 1)$ and $\omega_i > 0$, so the discrete check tests the algebra rather than the quadrature. Against adaptive quadrature the rule agrees to 5×10^{-15} relative error over the tested (γ, a) grid; at $\gamma = 0$, $S_I \equiv 1$ holds to 10^{-11} (largest deviation, at $\beta = 1.5$); (P2) analytic-versus-finite-difference

agreement is $8.6e-04$; the uniform bound (3) was verified on 540 parameter combinations with worst excess $0.0e{+}00$ and empirical tightness ratio 1.0000.

Appendix D. Bootstrap of γ

Pipes are resampled with replacement within the complete-history cohort (cluster bootstrap at the pipe level); for each replicate the panel is rebuilt, the aggregate retention recomputed and γ re-solved by bisection ($R = 60$ replicates, $\beta = 2.0$ and 2.91). The horizon interval is obtained by mapping the bootstrap endpoints of γ through $H(0.5;\gamma)$.